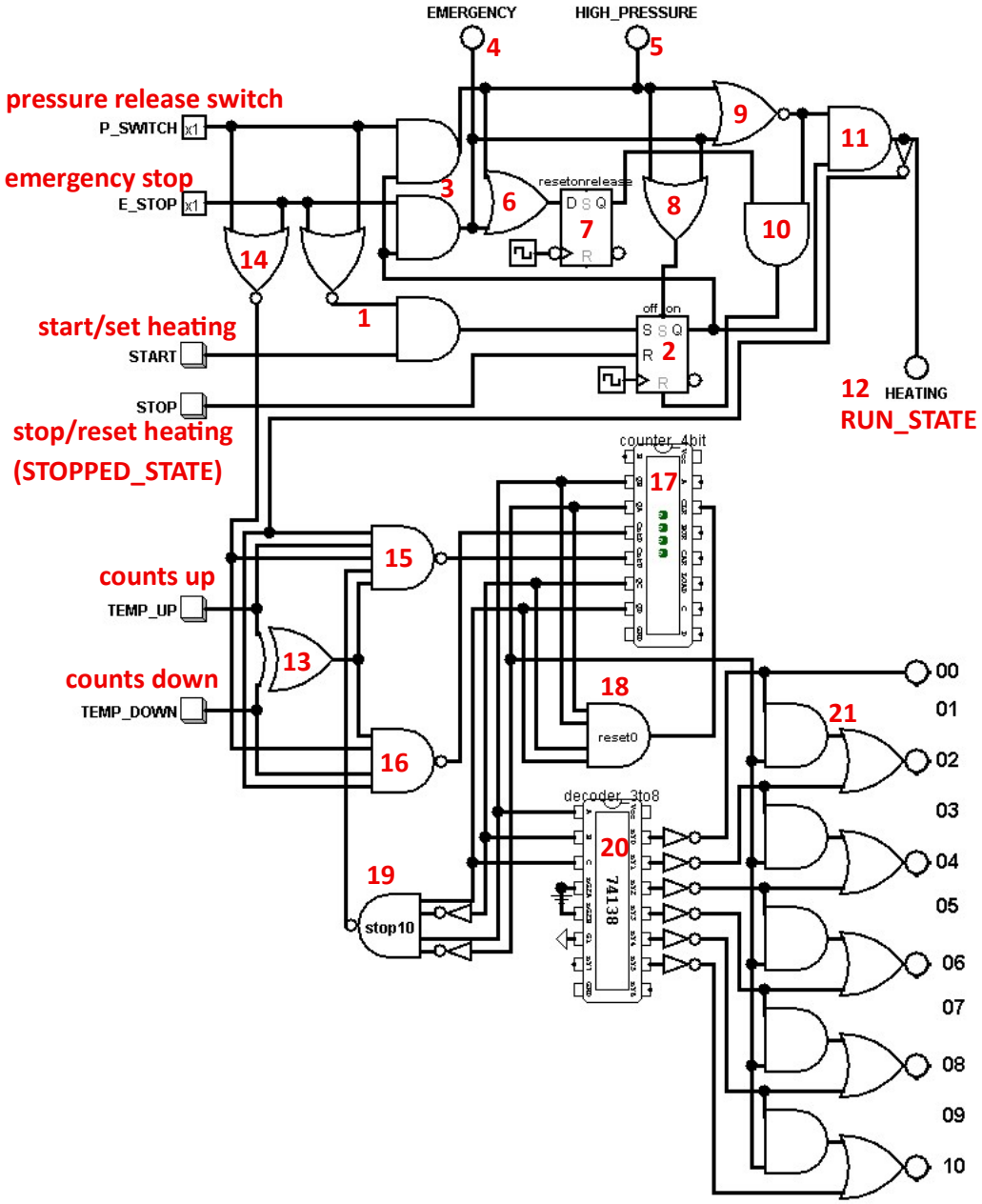


How the Logic Design Works

KEY

1. Only starts if P_SWITCH/E_STOP is off
2. Controls start(set)/stop(reset) determining Q output
3. P_SWITCH/E_STOP only activates while SR Q is ON
4. EMERGENCY LED ON = emergency state
5. HIGH_PRESSURE LED ON = high-pressure state
6. Carries any signal from P_SWITCH OR E_STOP to D Flip Flop
7. Stores 1 bit while P_SWITCH/E_STOP LEDs ON and holds it until falling edge (button release on 2nd press)
8. Any signal from P_SWITCH OR E_STOP carry to SR Flip Flop SET, disabling the SR from being RESET until the P_SWITCH OR E_STOP is deactivated
9. any signal passes and inverts, preventing heater from activating
10. because of D Flip Flop falling edge, there is a split moment the D Flip Flop Q and NOR (9) carry both signals through AND (10), resetting the SR Flip Flop once switches are released
11. SR Flip Flop Q is ON and emergency and/or high-pressure state are OFF (inverted from NOR (9))
12. HEATING LED is ON = RUN_STAT
13. Only TEMP_UP signal or TEMP_DOWN can pass to prevent both buttons being pressed simultaneously
14. Prevents TEMP_UP OR TEMP_DOWN from activating while E_STOP OR P_SWITCH is activated
15. for CntU requires all five inputs = ON
16. for CntD requires all four inputs = ON
17. stores 4-bits representing count value
18. resets at 0 to prevent counting decrement
19. blocks binary 1010 signal from counting above 10
20. converts 3 inputs to 8 outputs and NOT used to invert output signals
21. output AND QA output to OR gate to illuminate LEDs based on odd/even number e.g. 3 = 00 and 02 ON

Note: Both Flip Flops need a clock to operate